

Mass-Bound Time Scales and Interferometry

Why Holometer-Type Experiments Find No Discreteness Signal –
a Derivation from $\tilde{T} \cdot m = 1$

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Contents

A	Mass-Bound Time Scales and Interferometry	2
A.1	Background: Two Approaches to Space Quantisation	1
A.2	FFGFT Eigenfrequencies from $\tilde{T} \cdot m = 1$	1
	The Mass-Bound Time Scale	1
	External Inputs	1
	Numerical Values	1
	Comparison with Interferometer Frequencies	2
A.3	Lattice Dispersion Correction at the FFGFT Scale	2
A.4	The K_{frak} Correction and Its Measurability	2
A.5	The Central Statement on Interferometry Experiments	2
A.6	Relation to the D4 Lattice Argument	3

Chapter A

Mass-Bound Time Scales and Interferometry

Abstract

From the FFGFT fundamental relation $\tilde{T} \cdot m = 1$ follow mass-bound Compton frequencies $f_m = mc^2/h$ as the only physically grounded eigenfrequencies of a discrete substrate **[K]**. For all stable particles these lie at least 12 orders of magnitude above the measurement frequencies of current interferometers (MHz range). This yields a precise, derivable statement about interferometry experiments for the detection of space quantisation: not a design-induced blind spot but the mass-bound nature of the time scale $\tilde{T} = 1/m$ explains why Holometer-type experiments at their current sensitivity cannot find a discreteness signal **[K]**. A K_{frak} correction to the Compton frequency of $\Delta f_e/f_e = 100\xi_0 \approx 1.3\%$ acts only on the SI conversion; it is not a measurable spectral line, neither spectroscopically nor interferometrically **[K]**.

A.1 Background: Two Approaches to Space Quantisation

Experiments such as the Fermilab E-990 Holometer ($L = 40$ m, $f_c = c/(2L) \approx 3.75$ MHz) and the planned QUEST experiment ($L = 1.84$ m, $f_c \approx 81.5$ MHz) search for a quantisation signal of spacetime. The null results of these experiments have been interpreted from various quarters as evidence for or against specific discrete substrate models.

One concrete claim is that the interferometers are by design blind at exactly the frequency at which a discrete D4 lattice with lattice constant $a = \ell_P$ would resonate. This claim presupposes that the relevant substrate frequency is $f \sim c/(2L)$.

The FFGFT framework provides a more precise answer to the question of the relevant frequency scales — from first principles, without parameter fitting.

A.2 FFGFT Eigenfrequencies from $\tilde{T} \cdot m = 1$

The Mass-Bound Time Scale

[ASSUMPTION] In FFGFT, time is not an independent external quantity but is bound to mass by the fundamental relation $\tilde{T} \cdot m = 1$ (in natural units; in SI: $\tilde{T} \cdot m = \hbar/c^2$).

[K] Every massive particle therefore carries an intrinsic time scale:

$$\tilde{T}_m = \frac{\hbar}{mc^2}, \quad f_m = \frac{mc^2}{h}. \quad (\text{A.1})$$

This is the Compton frequency — not an additional postulate but a geometric consequence of the T^4 structure and the duality relation.

External Inputs

[ASSUMPTION] This document uses the following standard values as external inputs; they are declared here and not derived:

Quantity	Value	Source
m_e	9.1094×10^{-31} kg	PDG
m_μ	1.8836×10^{-28} kg	PDG
m_p	1.6726×10^{-27} kg	PDG
m_n	1.6749×10^{-27} kg	PDG
h	6.6261×10^{-34} J s	SI (derived from ξ)
c	2.9979×10^8 m/s	SI (derived from ξ)

Numerical Values

[K]

Compton frequencies of stable particles (SI)

Particle	$f_m = mc^2/h$	$R_m = \hbar/(mc^2)$
Electron	1.2356×10^{20} Hz	1.288×10^{-21} s
Muon	2.5548×10^{22} Hz	6.230×10^{-24} s
Proton	2.2687×10^{23} Hz	7.015×10^{-25} s
Neutron	2.2719×10^{23} Hz	7.005×10^{-25} s

Comparison with Interferometer Frequencies

[K]

$$\frac{f_e}{f_c^{\text{Holometer}}} = \frac{1.2356 \times 10^{20} \text{ Hz}}{3.747 \times 10^6 \text{ Hz}} \approx 3.3 \times 10^{13}, \quad (\text{A.2})$$

$$\frac{f_e}{f_c^{\text{QUEST}}} \approx 1.5 \times 10^{12}. \quad (\text{A.3})$$

The smallest mass-bound frequency (electron) thus lies 12 to 13 orders of magnitude above the measurement frequencies of current interferometers.

A.3 Lattice Dispersion Correction at the FFGFT Scale

[K] For a discrete substrate with lattice constant a , the lattice dispersion correction to linear dispersion is of order $(k \cdot a)^2$. At $k_1 = \pi/L$ this gives:

$$\left. \frac{\delta\omega}{\omega} \right|_{\text{lattice}} \sim \left(\frac{\pi a}{L} \right)^2. \quad (\text{A.4})$$

[K]

Lattice scale a	Holometer ($L = 40 \text{ m}$)	QUEST ($L = 1.84 \text{ m}$)
$\ell_P = 1.62 \times 10^{-35} \text{ m}$	$\sim 10^{-72}$	$\sim 10^{-70}$
$\hbar/(m_e c) = 3.86 \times 10^{-13} \text{ m}$	$\sim 10^{-27}$	$\sim 4 \times 10^{-25}$
$\hbar/(m_p c) = 2.10 \times 10^{-16} \text{ m}$	$\sim 3 \times 10^{-34}$	$\sim 10^{-31}$

Even at the Compton scale of the electron — which in the FFGFT framework is the *physical* fundamental scale, not ℓ_P — the lattice dispersion correction is $\sim 10^{-27}$, far beyond any measurable sensitivity.

A.4 The K_{frak} Correction and Its Measurability

[K] The fractal correction $K_{\text{frak}} = 1 - 100\xi_0$ with $\xi_0 = 4/30000$ acts exclusively on absolute SI values, not on the pullback isometry itself (Doc. 190, R96, R98). Applied to the Compton frequency of the electron:

$$\Delta f_e = f_e \cdot 100\xi_0 = f_e \cdot \frac{400}{30000} \approx 1.65 \times 10^{18} \text{ Hz}. \quad (\text{A.5})$$

This corresponds to a relative deviation of $100\xi_0 \approx 1.33\%$ from the Compton frequency. The value is arithmetically correct; but since K_{frak} acts only on absolute SI values, i.e. on the SI anchoring, the shift is not a directly measurable spectral line – it is accessible neither to spectroscopy (precision spectroscopy at $\sim 10^{20} \text{ Hz}$) nor to a Michelson interferometer in the MHz range.

A.5 The Central Statement on Interferometry Experiments

Conclusion: Why Holometer-type experiments find no signal

[K] In FFGFT the relevant frequency scale for space quantisation is not $f \sim c/(2\ell_p)$ (universal Planck frequency) but $f_m = mc^2/h$ (mass-bound Compton frequency). This scale differs for every particle and lies for all stable particles at least 12 orders of magnitude above the measurement frequencies of current interferometers.

It follows directly: Holometer-type experiments find nothing because the time scale of discreteness is not universal but mass-bound — not because a design-induced blind spot of the instrument lies exactly at the substrate resonance frequency.

The statement is falsifiable: an interferometer signal at frequency f would require a massless or ultralight field with $m = hf/c^2$. For $f = 3.75$ MHz this gives $m \approx 2.8 \times 10^{-44}$ kg $\approx 3 \times 10^{-14} m_e$ — no known particle.

A.6 Relation to the D4 Lattice Argument

[K] The D4 lattice with lattice constant $a = \ell_p$ has 24 nearest neighbours (8 of type $(\pm 1, 0, 0, 0)$ and 16 of type $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2})$). The dispersion relation of a scalar field on this lattice is:

$$\omega^2(\mathbf{k}) = \frac{c^2}{\ell_p^2} \sum_{\delta} (2 - 2 \cos(\mathbf{k} \cdot \delta \ell_p)). \quad (\text{A.6})$$

For $k_1 = \pi/L$ with $L = 40$ m one has $k_1 \ell_p \approx 1.3 \times 10^{-36}$, hence $k_1 \ell_p \ll 1$: the dispersion relation is exactly linear in this regime, identical to the continuum up to order $(k_1 \ell_p)^2 \sim 10^{-72}$. No special lattice resonance exists at $f_c = c/(2L)$ — the free spectral range $c/(2L)$ is an optical property of the interferometer, not a lattice property.

FFGFT replaces the D4 lattice argument with a stronger statement: not the lattice dispersion but the mass-bound time structure $\tilde{T} \cdot m = 1$ determines at which frequency a signal is to be expected.

Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

Verification script: `pruef_335_paul_frequenzen.py` (5/5 assertions).

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